An Introduction to Ergodicity Economics: Markets (II)

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Markets

6.5.3 Expected wealth and expected utility 86
6.6 From ergodicity transformation to wealth process and back 91
6.6.1 Itô setup 91
6.6.2 From ergodicity transformation to wealth process 92
6.6.3 From wealth process to ergodicity transformation . 95

7 Decisions in the real world 99
7.1 The St. Petersburg paradox 99
7.1.1 The St. Petersburg lottery 100
7.1.2 Expected-value theory – a paradox 101
7.1.3 Expected-utility theory – a resolution 101
7.1.4 Ergodicity economics – enter dynamics 102
7.2 The Copenhagen experiment 108
7.3 Insurance 111
7.4 A shipping insurance contract 114
7.4.1 Ergodicity economics solution 117
7.4.2 The classical solution of the insurance puzzle . . . 119

III Populations 121

8 People 123
8.1 Each one alone 123
8.1.1 Lognormal distribution 124
8.1.2 Two growth rates 125
8.1.3 Measuring inequality 129
8.1.4 Wealth condensation 132
8.1.5 Rescaled wealth 133
8.1.6 Power-law resemblance 134
8.2 Finite populations 136
8.2.1 How long is a long time? 136

9 Interactions 141
9.1 Cooperation 141
9.1.1 Farmers' fable 142
9.1.2 Model 143
9.2 Reallocation 150
9.2.1 The ergodic hypothesis in economics 151
9.2.2 Reallocating geometric Brownian motion 152
9.2.3 Positive reallocation 153
9.2.4 Negative reallocation 156
9.2.5 Fitting United States (US) wealth data 156

10 Markets 161
10.1 The portfolio selection problem 162

10.1.1 Riskless and risky assets 162
10.1.2 Leverage 163
10.1.3 Classical portfolio theory 166
10.1.4 Growth rate maximisation 170
10.2 Leverage efficiency 173
10.2.1 Relaxing the model 173
10.2.2 Efficiency 174
10.2.3 Stability 175
10.2.4 Prediction accuracy 177
10.3 Applications of leverage efficiency 180
10.3.1 A theory of noise 181
10.3.2 Solution of the equity premium puzzle 181
10.3.3 Central-bank interest rates 183
10.3.4 Fraud detection 184
10.4 Discussion 186

Acronyms 187

List of Symbols 189

Index 197

References 201

Markets

Economic agents can maximise their wealth over time by maximising the time-average growth rate of the wealth.

- What does that mean for the portfolio choice?
  - Optimal leverage
- What does that mean for the way markets organise?
  - Leverage efficiency

Optimal leverage

For a portfolio consisting of one risky asset (s) and one risk-free asset (r), the optimal share of wealth invested in the risky asset is:

$$\ell_{opt} = \frac{\mu_s - \mu_r}{\sigma_s^2} \tag{1}$$

Leverage efficiency:
Markets self-organise around

$$\ell_{opt} = 1$$

Leverage efficiency

Why $\ell_{opt} = 1$?

- Efficiency: It is impossible for an individual investor to beat the market simply by using leverage.
- Stability:
  - $\ell_{opt} > 1$: Everyone wants to borrow cash to buy stocks, but noone wants to lend money.
  - $\ell_{opt} < 0$: Everyone wants to borrow stocks and sell them to get cash, but noone wants to lend stocks.

Market simulation

The market consist of two agents, A and B,
one risky asset (s: GBM with exogenous parameters μ_s, σ_s)
and a debt contract (r: endogenous drift μ_r)

- A observes the risk-free rate, μ_r, and the stock dynamic, μ_s, σ_s
- A chooses its leverage, $\ell_A = \frac{\mu_s - \mu_r}{\sigma_s^2}$
- B accepts borrowing / lending at the risk-free rate according to A's needs.
- Portfolio outcomes are realised
- B adjusts the risk-free rate, μ_r

Market simulation

B's rate setting rule:

- If B's return was at least as good as A's during dt:
  - Do nothing
- If B's return was worse than A's during dt:
  - If A borrowed from B: Increase μ_r with $\Delta\mu dt$
  - If B borrowed from A: Decrease μ_r with $\Delta\mu dt$

A and B start out with the same wealth.

Predictions

Does $\ell_{opt} = 1$ hold for real markets?

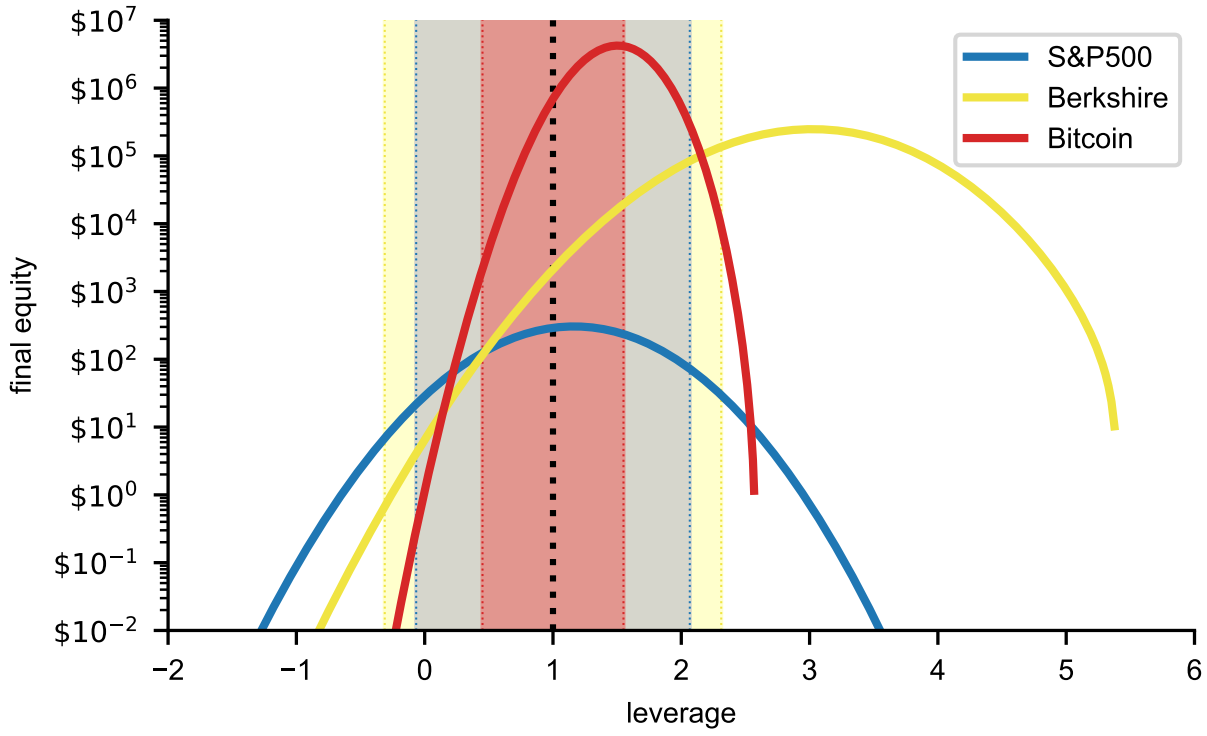

Figure 10.7 in Peters and Adamou (2025). Confidence bands indicate two standard errors of the prediction, where the standard error is $\sigma_{\ell_{opt}(\Delta t)} = \frac{1}{\sigma_s \Delta t^{1/2}}$.

Equity Premium Puzzle

Standard economic theory cannot explain the high premium on equity, but leverage efficiency can:

- $3 - 10\%$ in real life
- $0.1 - 0.5\%$ with standard theory (CCAPM)
- $0.4 - 6.4\%$ with optimal leverage ($s =$S&P500 and $r =$ FED overnight interbank rate)

Premium: difference in time average growth rate (ϵ)

$$\epsilon = \bar{g}_s - \bar{g}_r \tag{2}$$

$$= (\mu_s - \mu_r) - \frac{\sigma_s^2}{2} \tag{3}$$

$$= \frac{\mu_s - \mu_r}{2} \tag{4}$$

using leverage efficiency $\ell_{opt} = 1 \implies \mu_s - \mu_r = \sigma_s^2$

Summary

- For a risky and risk-free asset, the growth-rate optimal leverage is $\ell_{opt} = \frac{\mu_s - \mu_r}{\sigma_s^2}$
- Leverage efficient markets self-organise around $\ell_{opt} = 1$
- Leverage efficiency predicts a fixed relationship between drift, volatility and risk premium, supported by real life data

Thank you.